HEALTHCONNECT RWANDA

Business Insights Report

Healthcare Operations and Performance Analytics Dashboard

Source basis: Current MySQL SQL script and current six-page Power BI dashboard, interpreted together with the supplied Rwanda healthcare datasets and documented data-quality checks.

Executive findings

The analysis provides a strong enterprise view of patient demographics, recorded service activity, clinical workload, collections and patient experience. The most important management issue is not a lack of records but incomplete linkage: the supplied appointment parent table contains only 5,000 records while many related clinical and financial files reference additional appointment IDs. This limits facility- and department-level attribution even when source totals remain valid.

Executive KPI	Current result
Registered patients	20,000
Recorded appointments	5,000
Facilities represented	20
Doctors	350
Collected payments	RWF 8,389,454,556
Average satisfaction	2.99 / 5

Business implications

- Collections equal 90.12% of billed value. RWF 919.95 million remains outstanding across 11,887 bills; finance should reconcile balances and status rules before using billing-status labels for debt-recovery decisions.
- Patient demand is distributed broadly across provinces and visit types, but only the supplied 5,000 appointment records can support fully attributed facility, department and date analysis.
- Clinical activity is measurable through diagnosis, laboratory, prescription, admission and referral files, but unmatched appointment references must remain explicitly unknown rather than being imputed.
- Operational utilization cannot be fully assessed from static inventory alone: bed records are all marked Available and equipment files do not contain usage-hour history.

Implementation currently in use

The SQL implementation uses MySQL Workbench with the database name HEALTHCARE_ANALYTICS. The current script contains 40 analyses covering patient, appointment, clinical and financial questions. The Power BI file implements six report pages aligned to the project guide: Executive Dashboard, Patient Analytics, Clinical Analytics, Financial Dashboard, Operational Dashboard and Patient Experience Dashboard.

Patient and clinical insights

Patient profile

The patient register contains 10,090 male and 9,910 female patients, giving a near-even gender distribution. Average recorded age is 45.03 years, while 5,487 patients (27.44%) are above 65. A chronic condition is recorded for 14,929 patients (74.65%), led by asthma, hypertension and diabetes. These are registration characteristics and should not be interpreted as attendance rates.

Province of residence	Patients
Western	4,091
Eastern	4,067
Southern	3,967
Kigali City	3,952
Northern	3,923

Recorded service demand

Visit type	Appointments
Follow-up	1,694
Consultation	1,656
Emergency	1,650

Monthly appointment volume in the supplied file ranges from 400 in May to 440 in March. The small range supports a description of recorded activity, but the incomplete appointment parent file means it should not be treated as the full enterprise demand series.

Clinical activity

Top diagnosis	Records
Hypertension	1,059
Gastritis	1,017
Upper Respiratory Infection	1,004
Diabetes	985
Malaria	935

- Malaria is the most recorded laboratory test: 1,272 of 5,000 test records (25.44%).
- Amlodipine is the most prescribed medication: 1,068 of 5,000 prescription records.
- Average recorded admission duration is 5.49 days across 5,000 admission records.
- The referrals file contains 4,708 records; only 415 link to the supplied appointment parent table, so origin-facility analysis must preserve 4,293 referrals as unknown.

Financial performance

Financial measure	Amount (RWF)
Total billed	9,309,401,961
Collected payments	8,389,454,556
Outstanding billing balance	919,947,405
Insurance claim amount	4,881,863,293.88
Reported monthly revenue total	1,132,677,500

Collections and reconciliation

Recorded collections equal 90.12% of billed amounts. The average bill is approximately RWF 155,156.70. Every one of the 60,000 payment records matches a BillingID, while 42,036 bills are represented in the insurance claims file. Claims must remain separate from collections because an approved claim is a reimbursement request status, not proof that cash was received.

Payment method	Collected amount (RWF)
Card	2,115,489,446
Mobile Money	2,104,779,310
Insurance	2,094,324,407
Cash	2,074,861,393

- There are 21,008 approved and 21,028 pending insurance claims.
- Recorded billing status differs from a strict billed-versus-paid classification on 39,961 bills. Preserve source status and calculate balance separately.
- Only RWF 694,551,320 of payments can be attributed through the 5,000 available appointment parents, so facility and department revenue comparisons cover a subset unless an Unknown category is retained.

Revenue interpretation

The Revenue Summary provides 12 monthly values totaling RWF 1.133 billion and reports 60,809 appointments. Its accounting basis and coverage are not specified. Meanwhile, individual billing and payment records do not contain transaction dates. The two sources therefore should not be treated as directly comparable totals for the same period until their scope is confirmed.

Service-level revenue limitation: The current SQL script correctly excludes service-level revenue because the services table cannot be linked directly to appointments or billing by ServiceID. Joining only through DepartmentID would duplicate department payments and create misleading service revenue.

Operations and patient experience

Workforce and physical capacity

Staff role	People
Nurse	700
Doctor	350
Administrative Staff	80
Pharmacist	50
Laboratory Scientist	40
Radiographer	30

The combined staff register contains 1,250 unique staff IDs. The current Power BI Operational Dashboard compares specialty/department/facility staffing, patient volume by facility and appointment volume by department. These views support workload screening, but available appointment coverage should be stated whenever workload is calculated from appointment counts.

The source inventory contains 2,000 beds and 200 equipment records. All beds carry the status Available, which makes a status-based occupancy rate 0%. There are no dated occupancy records, usage hours or operating-time denominators, so true bed and equipment utilization over time cannot be calculated.

Patient experience

Feedback category	Records
Good	5,379
Poor	5,283
Excellent	5,204
Average	5,198

The feedback file contains 21,064 responses with an average rating of 2.99/5. Poor feedback represents 25.08% of responses. Only 1,756 feedback records (8.34%) link to available appointments, so hospital and department rankings based on appointment attributes represent a linked subset rather than the entire feedback file.

- The feedback data have no response date and no formal complaint flag.
- A negative-feedback trend can use the month of the linked appointment as a proxy only. This covers 422 Poor responses and should not be described as a dated complaint trend.
- Average rating should be shown together with response count so a small linked sample does not appear equivalent to broad patient coverage.

Current Power BI coverage

The uploaded PBIX contains all six required page names. Verified report objects include executive KPI cards and slicers; patient gender, age-group, province, chronic-condition and insurance/claim visuals; clinical diagnosis, test, prescription and referral visuals; financial monthly-revenue and collected-payment comparisons; operational staffing/volume/capacity visuals; and patient-experience rating, feedback and hospital-ranking visuals.

Data quality, risks and recommended actions

Appointment-parent coverage

The supplied Appointments table contains APT000001 through APT005000. Across the source set, 60,000 distinct appointment IDs are observed. The additional 55,000 referenced IDs do not have corresponding parent appointment details in the supplied appointment file. They should be retained in their source tables and left unattributed rather than being assigned invented patient, facility, department or date values.

Related dataset	Matched to supplied appointments	Total records
Diagnoses	5,000	5,000
Laboratory tests	2,272	5,000
Prescriptions	4,003	5,000
Admissions	586	5,000
Referrals	415	4,708
Patient feedback	1,756	21,064
Billing	5,000	60,000

Additional integrity risks

- Recorded FacilityID differs from the department-master association in 1,187 staff records and 4,744 appointments. Preserve recorded IDs until an authoritative facility/department master resolves the conflict.
- The transaction files do not provide dated billing, payment or claim events, so payment timeliness and financial time-series analysis are unavailable from transaction-level records.
- InsuranceID presence does not verify active coverage, and patient insurance codes do not provide a reliable crosswalk to named claim providers.
- The country master supports an RWF currency assumption, but the transaction tables do not contain explicit currency fields.

Recommended priorities

- Recover the complete appointment export, then rerun linkage, facility, department and time-based analyses.
- Reconcile the RWF 919.95 million outstanding balance and the 39,961 billing-status discrepancies using documented settlement rules and extraction dates.
- Confirm facility and department assignments against an authoritative master before using rankings for staffing or expansion decisions.
- Capture dated bed occupancy, equipment activity, complaint/feedback timestamps and service usage IDs to enable utilization, experience trends and service-level revenue.
- Keep claims, collected payments, billed value and reported monthly revenue as separate financial concepts until finance confirms their accounting scope.